

ASHISH PATEL

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EDUCATION

San Diego State University (San Diego, CA)

Aug 2024 - May 2026

M.S. Big Data Analytics, GPA: 4.0/4.0

Coursework: Enterprise Database Management, Big Data Tools & Methods, Machine Learning Engineering, Data Mining, Database Theory & Implementation

EXPERIENCE

Graduate Data Engineer | San Diego State University (San Diego, CA)

Sep 2024 - May 2026

- Designed an **AWS S3** research data lake with raw and curated zones, consolidating **4** heterogeneous source types (telemetry, mission logs, **REST APIs**, geospatial) into a single governed dataset for university research teams.
- Built reusable **Python, PySpark, and SQL** ingestion and schema-normalization pipelines to clean, standardize, and join all **4** sources, cutting downstream data-quality defects by **40%** through shared validation rules.
- Implemented **Great Expectations** validation suites and **Marquez/OpenLineage** lineage tracking, catching **schema drift**, null-rate spikes, and referential failures at ingest rather than in downstream analysis.
- Tuned **PySpark** jobs through partitioning, caching, and **Parquet** layout redesign and delivered self-serve **Streamlit** dashboards, reducing manual data preparation and reporting time by **80%** for research staff.

Data Engineer | Digit Insurance (India)

Jun 2022 - Jul 2024

- Engineered batch and near-real-time **ETL/ELT** pipelines in **Python, SQL, PySpark, and Kafka**, integrating claims, policy, billing, and REST API sources across **10M+** insurance records into a centralized **Databricks** lakehouse.
- Built **Delta Lake** medallion (Bronze/Silver/Gold) workflows using **incremental merge** patterns in place of full reloads, improving pipeline processing efficiency by **25%** and lowering daily cluster compute cost.
- Modeled claims, billing, and policy domains in **Snowflake** with **dbt**, delivering analytics-ready marts that cut data retrieval time by **50%** for underwriting and finance reporting.
- Orchestrated **Apache Airflow** DAGs with dependency management, retries, **SLA alerting**, and automated failure handling, improving pipeline reliability and on-time delivery of daily analytics refreshes.
- Embedded **dbt tests** and **PySpark** profiling for null, duplicate, referential-integrity, and schema-drift checks, and tuned **Spark** jobs through partitioning, caching, and execution-plan review to reduce shuffle overhead on large joins.
- Partnered with **claims operations** to automate recurring claims and finance reporting, eliminating **100+ hours** of manual work per month and delivering the daily curated **SLA dataset** that supported within-SLA claim settlement rising from **65% to 95%**.

SKILLS

Programming & Processing: Python, SQL, PySpark, Pandas, NumPy, Bash, CTEs, Window Functions

Data Engineering: Apache Spark, Databricks, Kafka, Apache Airflow, dbt, ETL/ELT, Batch & Near-Real-Time Pipelines, Incremental Loads

Cloud & Lakehouse: AWS S3, Azure Data Factory, Azure Data Lake, Snowflake, Delta Lake, Unity Catalog, Medallion Architecture

Warehousing & Modeling: PostgreSQL, Dimensional Modeling, Star Schema, Data Marts, Partitioning, Clustering, Query Optimization

Data Quality & DevOps: Great Expectations, Marquez/OpenLineage, dbt Tests, Data Validation, Profiling, Git, GitHub, Docker, CI/CD

Analytics: Power BI, Streamlit, DAX

PROJECTS

CureMatch: Clinical Trial Matching | *Python, SQL, SQLite, Next.js, TypeScript, Llama 3.3 70B, RAG*

[\[GitHub\]](#)

- Developed a **Python** ingestion pipeline loading **65K+** actively recruiting trials from the **ClinicalTrials.gov REST API** into **SQLite** (**411K+** locations, **236K+** interventions), plus an **RxNorm** and regex parser that extracts medication exclusions, lab thresholds, and ECOG ranges in **86 seconds**.
- Designed a deterministic **6-criterion** scoring engine whose **SQL** pre-filter narrows **65K** trials to **~1,500** candidates in milliseconds, with a **RAG** layer on **Llama 3.3 70B** that explains match verdicts in plain English without altering match decisions.

Snowflake dbt Airflow Data Pipeline | *Snowflake, dbt, Apache Airflow, Astronomer Cosmos, AWS S3, Docker*

[\[GitHub\]](#)

- Engineered an **S3-to-Snowflake ELT** pipeline for Airbnb booking, listing, and host data, orchestrated as **Apache Airflow** DAGs via **Astronomer Cosmos** on a containerized **Docker** deployment.
- Built **8 dbt models** and **3 SCD Type 2 snapshots** with validation tests, producing history-tracked dimensional and **OBT** outputs for repeatable daily warehouse refreshes.

Airline Operations Intelligence | *PostgreSQL, SQL, Dimensional Modeling, Power BI, DAX*

[\[GitHub\]](#)

- Modeled **8.4M+** U.S. domestic flight records into a **PostgreSQL star schema** and developed **5** analytical SQL workloads using **CTEs**, **window functions**, **LAG**, **RANK**, and **NTILE**.
- Identified **76.5%** of delay minutes as operationally controllable with late-aircraft propagation at **39.4%**, and built a **Power BI** dashboard with **DAX** KPIs and a what-if model sizing **5-30%** delay reductions across **17 carriers**.